

Sensitivity Analysis of Azeotropic Distillation Process

Method

Temperature was varied from 60°C to 90°C while keeping feed composition and pressure constant. At each temperature, vapor pressures were calculated using the Antoine equation and vapor composition was estimated. Ethanol mole fraction in the vapor phase was recorded as the purity.

Results

Temperature (°C)	Ethanol Purity (Mole Fraction)
60	0.4431
62	0.4443
64	0.4455
66	0.4466
68	0.4477
70	0.4488
72	0.4498
74	0.4508
76	0.4518
78	0.4527
80	0.4536
82	0.4545
84	0.4554
86	0.4562
88	0.4570
90	0.4578

Table : Effect of Temperature on Ethanol Purity

Graph

The graph of Temperature vs Ethanol Purity is plotted. It shows that as temperature increases, ethanol purity also increases.

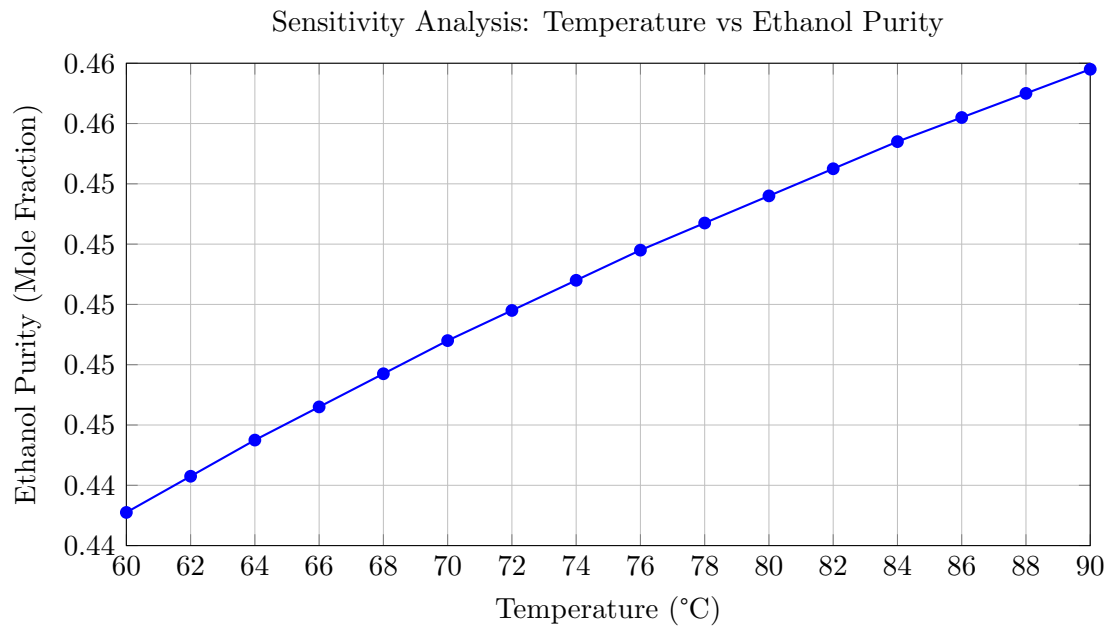


Figure : Variation of ethanol purity with temperature

Observation

It is observed that ethanol purity increases with temperature. This is because higher temperatures increase vapor pressure and relative volatility, enhancing the separation efficiency in the distillation process.

Code Snippet

```
import numpy as np
import matplotlib.pyplot as plt

# Antoine function
def antoine(T, A, B, C):
    return 10**(A - B/(C + T))

# Constants (correct values)
A = [8.20417, 8.07131, 6.8493]          # Ethanol, Water, Cyclohexane
B = [1642.89, 1730.63, 1203.835]
C = [230.3, 233.426, 222.65]

P = 760 # mmHg

# Assume composition (can keep fixed)
x = np.array([0.4, 0.5, 0.1])
```

```

T_list = []
purity = []

for T in range(60, 92, 2):

    Psat = np.array([
        antoine(T, A[0], B[0], C[0]),
        antoine(T, A[1], B[1], C[1]),
        antoine(T, A[2], B[2], C[2])
    ])

    # Assume gamma = 1 (simplified for sensitivity)
    gamma = np.array([1, 1, 1])

    y = (x * gamma * Psat) / P
    y = y / sum(y)

    T_list.append(T)
    purity.append(y[0])    # ethanol

    print(f"T = {T} C      Ethanol purity = {y[0]:.4f}")

# Plot
plt.plot(T_list, purity, marker='o')
plt.xlabel("Temperature ( C )")
plt.ylabel("Ethanol Purity")
plt.title("Sensitivity Analysis: Temperature vs Ethanol Purity")
plt.grid()
plt.show()

```